M3503 WEEK 1 2021

First Order Ordinary Differential Equations

These are equations of the form

$$\frac{dy}{dx} = f(x,y)$$

Some classes of first order ordinary differential equations include,

1. Separable Equations

These are equations with the form

$$\frac{dy}{dx} = g(x)\,h(y)$$

$$\Rightarrow \int \frac{1}{h(y)}\,dy = \int g(x)\,dx$$

$$\Rightarrow H(y) = G(x) + C$$

Examples

Solve (a) $y' = \frac{-x}{y}$

(b) $e^x y' + x[1 + y^2] = 0 \quad ; \quad y(-1) = 0$

Soln (a) $y' = \frac{-x}{y}$

$$\Rightarrow y\,dy = -x\,dx$$

$$\Rightarrow \tfrac{1}{2}y^2 = -\tfrac{1}{2}x^2 + C_1$$

$$\Rightarrow y^2 + x^2 = C_2$$

(b) $e^x y' + x(1+y^2) = 0 \quad ; \quad y(-1) = 0$

$$\Rightarrow \frac{dy}{1+y^2} = -xe^{-x}dx$$

$$\Rightarrow \tan^{-1} y = xe^{-x} + e^{-x} + C$$

$$\tan^{-1}(0) = -e + e + C \Rightarrow C = 0$$

$$\therefore \tan^{-1} y = xe^{-x} + e^{-x}$$

2. Linear First Order Equations

These are equations which can be expressed in the form

$$y' + P(x)y = f(x) \quad \text{—— ①}$$

Methods For Solving First Order Linear Equations

- Express the linear equation in standard form (as in eq^n ①)
- Calculate the integrating factor u given by $u = \exp(\int P(x)dx)$, with constant of integration taken as zero.
- Multiply the equation in standard form by u to get $uy' + Puy = uf$

$$\Rightarrow \frac{d}{dx}(uy) = uf$$

$$\Rightarrow uy = \int uf\,dx + C$$

since $u = \exp(\int P dx)$

$\Rightarrow \ln u = \int P dx$

$\Rightarrow \frac{1}{u} du = P dx$

$\Rightarrow \frac{du}{dx} = uP$

Examples

solve

1 (a) $x\frac{dy}{dx} + 3(y+x^2) = \frac{\sin x}{x}$

(b) $\frac{dy}{dt} = 50e^{-10t} - ky$

soln

(a) $x\frac{dy}{dx} + 3(y+x^2) = \frac{\sin x}{x}$

$\Rightarrow \frac{dy}{dx} \quad \frac{3y}{x} = \frac{\sin x}{x^2} - 3x$

$IF = u = \exp\left(\int \frac{3}{x}\,dx\right)$

$= \exp(3\ln x)$

$= x^3$

$x^3\frac{dy}{dx} + 3x^2y = x\sin x - 3x^4$

$\Rightarrow \frac{d}{dx}(yx^3) = x\sin x - 3x^4$

$\Rightarrow yx^3 = -x\cos x + \sin x - \frac{3}{5}x^5 + C$

(b) $\frac{dy}{dt} = 50e^{-10t} - ky$

$\Rightarrow \frac{dy}{dt} + ky = 50e^{-10t}$

$u = \exp\int k\,dt$

$= e^{kt}$

$$\frac{d}{dt}(ye^{kt}) = 50e^{(k-10)t}$$

$$\Rightarrow ye^{kt} = \frac{50}{k-10}e^{(k-10)t} + C$$

3. Exact Equations

The differential equation $M(x,y)\,dx + N(x,y)\,dy = 0$ ① is exact if there exists a function $f(x,y)$ such that $f_x = M(x,y)$ and $f_y = N(x,y)$.

Test For Exactness

Equation ① is exact if the condition

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \forall (x,y)$$ holds

Methods For Solving Exact Equations

- Test for exactness, i.e check if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
- If the equation is exact, then the exists f such that $\frac{\partial f}{\partial x} = M(x,y)$

$$\Rightarrow f = \int M(x,y)\,dx + h(y) \quad \text{——} \circledast$$

- To find $h(y)$, differentiate f in $\circledast$ with respect to y and equate it to N, since $f_y = N$.
- Solve for $h'(y)$ and integrate to get $h(y)$, substitute it into $\circledast$

The general solution is given by

$$f(x,y) = C$$

Examples

Solve

(a) $(2xy+3)\,dx + (x^2-1)\,dy = 0$

(b) $(y\cos x + 2xe^y) + (\sin x + x^2e^y - 1)\,y' = 0$

Sol^n

(a) let $M = 2xy+3$, $N = x^2-1$

$\Rightarrow M_y = 2x$, $N_x = 2x$

$M_y = N_x$.

There exists a function f such that $f_x = M$ and $f_y = N$

let $f_x = 2xy + 3$

$\Rightarrow f = x^2y + 3x + h(y)$

$\Rightarrow f_y = x^2 + h'(y) = x^2 - 1$

$\Rightarrow h'(y) = -1$

$\Rightarrow h(y) = -y + k$

$\therefore x^2y + 3x - y = C$ is the general sol^n

(b) let $M = y\cos x + 2xe^y$, $N = \sin x + x^2e^y - 1$

$\Rightarrow M_y = \cos x + 2xe^y$, $N_x = \cos x + 2xe^y$

$M_y = N_x$

There exists a function f such that $f_x = M$ and $f_y = N$

let $f_y = \sin x + x^2e^y - 1$

$\Rightarrow f = y\sin x + x^2e^y - y + h(x)$

$\Rightarrow f_x = y\cos x + 2xe^y + h'(x) = y\cos x + 2xe^y$

$\Rightarrow h'(x) = 0$

$\Rightarrow h(x) = k$

$\therefore y\sin x + x^2e^y - y = C$

A Non-Exact Equation Made Exact

If $M(x,y)\,dx + N(x,y)\,dy = 0$ is not exact but

$$\mu(x,y)M(x,y)\,dx + \mu(x,y)N(x,y)\,dy = 0 \quad \text{—①}$$

is exact, then μ is the integrating factor of $M\,dx + N\,dy = 0$

Since ① is exact, we have

$$\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N)$$

$$\Rightarrow \mu\frac{\partial M}{\partial y} + M\frac{\partial \mu}{\partial y} = \mu\frac{\partial N}{\partial x} + N\frac{\partial \mu}{\partial x}$$

$$\Rightarrow \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right)\mu = N\frac{\partial \mu}{\partial x} - M\frac{\partial \mu}{\partial y} \quad \text{—②}$$

Solving ② for μ can be challenging except for $\mu = \mu(x)$ and $\mu = \mu(y)$

Let's assume that $\mu = \mu(x)$, then ② becomes

$$\left(\frac{M_y - N_x}{N}\right)\mu = \frac{\partial \mu}{\partial x}, \quad \text{where } \frac{M_y - N_x}{N} = f(x)$$

$$\Rightarrow \frac{1}{\mu}\,d\mu = \left(\frac{M_y - N_x}{N}\right)dx$$

$$\Rightarrow \ln\mu = \int \frac{M_y - N_x}{N}\,dx$$

$$\Rightarrow \mu = \exp\left(\int \frac{M_y - N_x}{N}\,dx\right), \quad \frac{M_y - N_x}{N} = f(x)$$

Similarly, if $\mu = \mu(y)$, then

$$\mu = \exp\left(\frac{N_x - M_y}{M}\,dy\right), \quad \frac{N_x - M_y}{M} = f(y)$$

Examples

Solve

① $xy\,dx + (2x^2 + 3y^2 - 20)\,dy = 0$

② $(2x^2 + y)\,dx + (x^2y - x)\,dy = 0$

sol^n let $M = xy$, $N = 2x^2 + 3y^2 - 20$

$M_y = x$, $N_x = 4x$

$$\frac{N_x - M_y}{M} = \frac{4x - x}{xy} = \frac{3}{y} = f(y)$$

thus we have the integrating factor as $e^{\int 3/y\,dy} = y^3$

so; $xy^4\,dx + (2x^2y^3 + 3y^5 - 20y^3)\,dy = 0$ is exact

since $4xy^3 = 4xy^3 \Rightarrow M_y = N_x$

There exists f such that $f_x = M$ and $f_y = N$

$f_x = xy^4$

$\Rightarrow f = \frac{1}{2}x^2y^4 + h(y)$

$\Rightarrow f_y = 2x^2y^3 + h'(y) = (2x^2y^3 + 3y^5 - 20y^3)$

$\Rightarrow h'(y) = 3y^5 - 20y^3$

$\Rightarrow h(y) = \frac{1}{2}y^6 - 5y^4 + k$

$\therefore \frac{1}{2}x^2y^4 + \frac{1}{2}y^6 - 5y^4 = C$

② Let $M = 2x^2 + y$, $N = x^2y - x$

$M_y = 1$, $N_x = 2xy - 1$

$$\frac{M_y - N_x}{N} = \frac{1 - 2xy + 1}{x^2y - x} = \frac{-2(xy - 1)}{x(xy - 1)} = -\frac{2}{x} = f(x)$$

Integrating factor $= \exp\int \frac{-2}{x}\,dx$

$= \exp(\ln x^{-2})$

$= x^{-2}$

So $(2 + yx^{-2})\,dx + (y - x^{-1})\,dy = 0$ is exact since

$x^{-2} = x^{-2} \Rightarrow M_y = N_x$

There exists f such that $f_x = M$ and $f_y = N$

Let $f_x = 2 + yx^{-2}$

$\Rightarrow f = 2x - yx^{-1} + h(y)$

$\Rightarrow f_y = -x^{-1} + h'(y) = y - x^{-1}$

$\Rightarrow h'(y) = y$

$\Rightarrow h = \frac{1}{2}y^2 + k$

$\therefore 2x - yx^{-1} + \frac{1}{2}y^2 = C$

4. Homogeneous Equations

If $f(kx, ky) = k^n f(x,y)$, then $f(x,y)$ is a homogeneous function of degree n.

A differential equation $M(x,y)\,dx + N(x,y)\,dy = 0$ is a homogeneous equation if M and N are homogeneous functions of the same degree.

To solve a homogeneous equation, we make the substitution $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$.

This substitution transforms a homogeneous equation into a variable separable equation.

Examples, solve

$$(xy + y^2 + x^2)dx - x^2 dy = 0$$

Soln

$$(xy + y^2 + x^2)dx - x^2 dy = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{xy + y^2 + x^2}{x^2}$$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Substition gives

$$v + x\frac{dv}{dx} = \frac{x^2 v + x^2 v^2 + x^2}{x^2}$$

$$\Rightarrow v + x\frac{dv}{dx} = v + v^2 + 1$$

$$\Rightarrow \frac{1}{v^2+1}\,dv = \frac{1}{x}\,dx$$

$$\Rightarrow \tan^{-1}(v) = \ln(x) + C$$

$$\Rightarrow \tan^{-1}(y/x) = \ln x + C$$

$$\Rightarrow y = x\tan(\ln x + C)$$

5. Differential Equations Of The Form

$$\frac{dy}{dx} = f(ax+by)$$

To solve the equation, we make the substitution $z = ax+by$, which transforms the equation into a separable equation.

Examples

Solve

① $\frac{dy}{dx} = \sqrt{x+y} - 1$

② $\frac{dy}{dx} = y - x - 1 + (x-y+2)^{-1}$

Soln

① let $z = x+y$

$\Rightarrow \frac{dz}{dx} = 1 + \frac{dy}{dx}$

Substition gives

$$\frac{dz}{dx} - 1 = \sqrt{z} - 1$$

$\Rightarrow z^{-\frac{1}{2}} dz = dx$

$\Rightarrow 2\sqrt{z} = x + C$

$\Rightarrow 2\sqrt{x+y} = x + C$

② $\frac{dy}{dx} = y - x - 1 + (x - y + 2)^{-1}$

let $z = -y + x$

$\Rightarrow \frac{dz}{dx} = -\frac{dy}{dx} + 1$

Substitution gives

$1 - \frac{dz}{dx} = -z - 1 + (z+2)^{-1}$

$\Rightarrow \frac{dz}{dx} = (2+z) - (2+z)^{-1}$

$\Rightarrow \frac{dz}{dx} = \frac{(2+z)^2 - 1}{2+z}$

$\Rightarrow \frac{z+2}{(z+2)^2 - 1} = dx$

$\Rightarrow \frac{1}{2} \ln\left[(z+2)^2 - 1\right] = x + C$

$\Rightarrow \frac{1}{2} \ln\left[(x-y+2)^2 - 1\right] = x + C$

let $u = (z+2)^2 - 1$

$\Rightarrow du = 2(z+2)\,dz$

$\Rightarrow \frac{1}{2} du = (z+2)\,dz$

$\frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln u$